

AMIRHOSSEIN DABIRIAGHDAM

University of British Columbia, Vancouver, Canada

✉ Email 🌐 Personal Website 🎓 Google Scholar 🐙 GitHub 🔗 LinkedIn

RESEARCH INTERESTS

- Large Language Models
- Vision-Language Models
- LLM Agents
- World Models

EDUCATION

University of British Columbia (UBC)

- **Ph.D.** in Electrical and Computer Engineering Sep. 2025 - Present
GPA: 98/100 (A+)

- **M.A.Sc.** in Electrical and Computer Engineering Sep. 2023 - Aug. 2025
Thesis: “Combating Disinformation in the Age of Generative AI”
GPA: 96.3/100 (A+) (Thesis Grade: 100/100)

University of Tehran (UT) Sep. 2018 - Jul. 2023

- **B.Sc.** in Electrical and Computer Engineering
Thesis: “An Analysis of Forgettable Examples Extracted During Multilingual Models Training”
GPA: 19.21/20 (Ranked 2nd among 120 students)

SELECTED PUBLICATIONS

Peer-Reviewed Publications

- SimMark: A Robust Sentence-Level Similarity-Based Watermarking Algorithm for LLMs
Amirhossein Dabiriaghdam, Lele Wang
EMNLP 2025 Main Conference Oral (325/8174 = 3.98%), Suzhou, China, November 2025.
- SOI Matters: Analyzing Multi-Setting Training Dynamics in Pretrained Language Models via Subsets of Interest
Shayan Vassef*, **Amirhossein Dabiriaghdam***, Mohammadreza Bakhtiari*, Yadollah Yaghoobzadeh
MRL 2025 Workshop @ EMNLP 2025, Suzhou, China, November 2025.
- BCAmirs at SemEval-2024 Task 4: Beyond Words: A Multimodal and Multilingual Exploration of Persuasion in Memes
*Amirhossein Abaskohi**, **Amirhossein Dabiriaghdam***, Lele Wang, Giuseppe Carenini (*Equal Contribution)
SemEval-2024 Workshop @ NAACL 2024, Mexico City, Mexico, June 2024.
- Targeted Adversarial Attacks against Neural Machine Translation
Sahar Sadrizadeh, **Amirhossein Dabiriaghdam**, Ljiljana Dolamic, Pascal Frossard
IEEE ICASSP, Rhodes Island, Greece, June 2023.

Preprints

- VAMPS: Visual-Assisted Mathematical Problem Solving Benchmark
Amirhossein Dabiriaghdam, Shayan Vassef, Mohammadreza Bakhtiari, Yasamin Medghalchi, et al.
arXiv preprint arXiv:2606.04244, 2026.
- UnpredictaBench: A Benchmark for Evaluating Distributional Randomness in LLMs
*Amirhossein Abaskohi**, **Amirhossein Dabiriaghdam***, Liang Luo, Ellie Dingqiao Wen, Lele Wang, Giuseppe Carenini, Peter West
arXiv preprint arXiv:2606.06622, 2026. (*Equal Contribution)
- When Minor Edits Matter: LLM-Driven Prompt Attack for Medical VLM Robustness in Ultrasound
Yasamin Medghalchi, Milad Yazdani, **Amirhossein Dabiriaghdam**, Moein Heidari, et al.
arXiv preprint arXiv:2603.21047, 2026.

RESEARCH EXPERIENCE

- **Graduate Research Assistant** Sep. 2023 - Present
University of British Columbia Vancouver, Canada

- **Ph.D.:** Developing *agentic multimodal LLMs for reasoning and decision-making in interactive visual environments*, with a focus on (visual) *tool use*, *anticipatory reasoning with world models*, and *multi-agent collaboration*. Building *benchmarks and evaluation frameworks* for *multimodal reasoning* and *distributional generation* to characterize model capabilities, failure modes, and generalization.

Advisors: Prof. Giuseppe Carenini and Prof. Lele Wang.

- **M.A.Sc.:** Developed *SimMark*, a robust sentence-level similarity-based *watermarking algorithm* for LLM-generated text (EMNLP 2025), and studied multimodal and multilingual persuasion analysis.

- **Research Internship - EPFL Excellence in Engineering (E3)** *Jul. 2022 - Sep. 2022*
Signal Processing Laboratory 4 (LTS4), EPFL *Lausanne, Switzerland*

- Worked on the targeted *adversarial attacks* against neural machine translation models (ICASSP 2023).
Advisor: Prof. Pascal Frossard.

- **Summer Internship** *Jul. 2021 - Sep. 2021*
HARA.ai Co *Tehran, Iran*

- Developed a chatbot *natural language understanding unit* using state-of-the-art deep learning models.
Advisor: Prof. Reshad Hosseini.

HONORS & AWARDS

- Awarded 4 Year Doctoral Fellowship (4YF) from UBC. *2025 - 2029*
- Received GSI Award from UBC. *2023 - 2025*
- Received the Faculty of Engineering (FOE) Award from UT. *2019*
- Ranked 537th among about 150000 participants in the Nationwide University Entrance Exam. *2018*
- Kyokushin Karate Black Belt holder and member of IKO Kyokushinkaikan. *2008 - Present*

TEACHING EXPERIENCE

- **Graduate Academic Assistant** *Jun. 2026 - Sep. 2026*
University of British Columbia *Vancouver, Canada*

- AI 100: Introduction to Artificial Intelligence
Contributed to the design and development of course content, assessments, lab activities, instructional materials, and software infrastructure for a broad-audience introductory AI course. Collaborated with Profs. Kevin Leyton-Brown, Giulia Toti, and Elham Khoda, along with other AI 100 faculty.

- **Graduate Teaching Assistant** *Jan. 2024 - Present*
University of British Columbia *Vancouver, Canada*

- AI 422: Intelligent Systems
- AI 322: Foundations of Artificial Intelligence
- CPEN 311: Digital Systems Design

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, MATLAB, Shell scripting
- **ML / DL Frameworks:** Hugging Face (Transformers, Datasets, Diffusers), PyTorch, NumPy, Scikit-learn
- **Tools:** Linux, Git, L^AT_EX, Slurm, vLLM, DeepSpeed

LANGUAGES

- **English** Full professional proficiency
- **French** Elementary proficiency
- **Turkish (Azari)** Native (Bilingual proficiency)
- **Persian** Native (Bilingual proficiency)

SERVICE & LEADERSHIP

- **Graduate Student Representative** *2025 - 2026*
ECE Graduate Student Association (ECEGSA), UBC
- **Reviewer** *2025*
International Symposium on Topics in Coding (ISTC 2025)
- **Staff Volunteer** *2024*
18th Canadian Workshop on Information Theory (CWIT 2024)